

Robinson George Arysseril

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EDUCATION

Vimal Jyothi Engineering College

B.Tech in Artificial Intelligence and Data Science CGPA: 9.46 / 10

- Minor in Electronics and Communications

Chemperi, Kerala

Sep. 2024 – May 2028

TECHNICAL SKILLS

Systems & Embedded: C/C++, Rust, Zig, Xtensa Assembly

Low-Level Concepts: SPI, UART, GPIO, FAT32, MBR, Memory-Mapped I/O, Boot Stages, DMA, Interrupt Handling

General Purpose: Python, Java, JavaScript, TypeScript, MySQL

Developer Tools: Git, Make

PROJECTS

Bare Metal ESP32 Kernel 🐛 | *C, Xtensa Assembly, Python*

Dec. 2025 – Present

- Designed a two-stage boot process where a flash-resident bootloader reads a kernel binary from SD card into DRAM, then copies it to IRAM via a custom Xtensa assembly trampoline before execution
- Register-level SPI driver built from ESP32 TRM documentation, covering GPIO matrix routing, IO MUX configuration, and full SD card SPI mode initialization
- FAT32 filesystem driver supporting file read, write, delete, and directory listing with MBR partition table parsing and FAT cluster chain traversal
- GPIO and UART hardware abstraction layers enabling real-time GPIO control and bidirectional serial communication with the host at runtime
- Interactive kernel shell and domain-specific script interpreter that loads programs from SD card and executes timed GPIO sequences, enabling runtime reprogramming without reflashing

Various Emulators | *Rust, C++, Zig*

Oct. 2024 – Present

- **Game Boy (DMG) 🐛:** Cycle-accurate emulator implementing CPU (Sharp LR35902), PPU, memory controller, and MBC1/MBC3 cartridge mappers with scanline-based rendering, tile caching, sprite compositing, window layer support, and OAM DMA timing
- **Intel 8080 / Space Invaders 🐛:** CPU emulator with complete instruction set support, accurate flag operations, and memory-mapped I/O with interrupt-driven timing synchronized to cycle counts, reproducing alternating VBLANK interrupts at hardware-accurate intervals
- **XO-CHIP 🐛:** Extended instruction set in Zig supporting dual display planes, 64KB memory addressing via 16-bit index register extension, and CHIP-8/SCHIP compatibility including low/high resolution modes, scrolling, and persistent flag storage
- **CHIP-8 🐛🐛:** Interpreter in both Rust and C++ with full opcode coverage, stack operations, timers, input handling, and a framebuffer graphics system with sprite wrapping and collision detection

Chrultrabook Tools 🐛 | *Rust, Tauri, Angular, TypeScript*

May 2023 – Present

- Cross-platform desktop application enabling custom fan control, hardware diagnostics, keyboard backlight management, and key remapping for Chromebooks running alternate operating systems, with over 45,000 downloads
- Rust backend for direct hardware access, low-level sensor polling, and fan curve control across Windows, macOS, and Linux
- Modular architecture supporting multiple Chromebook hardware generations with minimal code changes for new device integration

One-Click Driver Installer & Hardware Database 🐛🐛 | *Rust, JSON, SQL*

Sep. 2024 – Oct. 2024

- Rust CLI tool automating driver detection and installation for alternate OS deployments on Chromebook hardware; over 5,000 downloads
- JSON and SQL database cataloging 200+ Chromebook models with chipset information and driver mappings; dynamic runtime fetch eliminates rebuilds on new model additions

EXPERIENCE

Open Source Contributor

May 2023 – Present

Chrultrabook Project

Remote

- Maintained Chrultrabook Tools, a cross-platform system utility serving users running alternate operating systems on Chromebook hardware
- Provided technical support and troubleshooting for community members transitioning Chromebooks to Linux, Windows, and other operating systems
- Collaborated with project maintainers to improve documentation, hardware compatibility databases, and user-facing tooling